



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Product Design with Autodesk Inventor

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Conducted by

Tertiary Infotech Academy Pte Ltd

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
8.0	16 July 2026	Legacy trainer deck and existing delivery materials.	Tertiary Infotech Academy
9.0	20 August 2026	Rebuilt visual courseware, detailed Lesson Plan, eight progressive labs, native charts and aligned assessments.	Dr. Alfred Ang

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Course Overview

A two-day, 16-hour WSQ course aligned to Computer-Aided Design (PRE-DES-4036-1.1). Learners progress from design intent and parametric parts through assemblies, weldments, manufacturing drawings, presentation, automation and final design review.

Learning Outcomes

- LO1: Conceptualise and generate machine parts using Autodesk Inventor.
- LO2: Generate CAD models using Autodesk Inventor.
- LO3: Produce assemblies using Autodesk Inventor.
- LO4: Connect and prepare assemblies for welding.
- LO5: Produce detailed drawings for machine parts.
- LO6: Review final designs for automation and improvement.

Daily Schedule

Day 1

Time	Duration	Topic / Activity	Method	Slides
9:00–9:30	30 min	Administration, digital attendance, introductions and assessment briefing	Facilitation	Slides 1–15
9:30–10:30	60 min	Topic 1 — Generate Machine Parts	Concepts, demonstration and review	Slides 16–28
10:30–10:40	10 min	Tea break	Break	—
10:40–12:15	95 min	Lab 1 — Design Brief to Parametric Base Part	Hands-on lab	Slide 27
12:15–1:15	60 min	Lunch break	Break	—
1:15–2:25	70 min	Topic 2 — Generate CAD Models	Concepts, demonstration and review	Slides 29–41
2:25–3:45	80 min	Lab 2 — Advanced Part, Sheet Metal and Orthographic Review	Hands-on lab	Slide 40
3:45–3:55	10 min	Tea break	Break	—
3:55–4:45	50 min	Topic 3 — Assembly Modeling	Concepts and demonstration	Slides 42–55
4:45–5:35	50 min	Labs 3–4 — Bottom-up and top-down assemblies	Hands-on lab	Slides 53 and 54

5:35–6:00	25 min	Day 1 integration, evidence check and PM attendance	Review and Q&A	Slide 55
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Day 2

Time	Duration	Topic / Activity	Method	Slides
9:00–9:15	15 min	Day 1 recap and AM digital attendance	Review	—
9:15–10:30	75 min	Topic 4 and Lab 5 — Weldments	Concepts, demonstration and lab	Slides 56–68 · Lab 67
10:30–10:40	10 min	Tea break	Break	—
10:40–12:15	95 min	Topic 5 and Lab 6 — Deliverables and GD&T	Concepts, demonstration and lab	Slides 69–81 · Lab 80
12:15–1:15	60 min	Lunch break	Break	—
1:15–2:25	70 min	Topic 6 and Lab 7 — Presentation and animation	Concepts, demonstration and lab	Slides 82–95 · Lab 93
2:25–2:35	10 min	Tea break	Break	—
2:35–3:20	45 min	Lab 8, final review and course summary	Hands-on lab and review	Slide 94 onward
3:20–3:30	10 min	Assessment briefing and digital attendance	Assessment administration	—
3:30–4:30	60 min	Written Assessment (WA-SAQ)	Open-book assessment	—
4:30–6:00	90 min	Practical Performance Assessment (PP)	Individual practical assessment	—

Topic-by-Topic Delivery Notes

Topic 1 — Generate Machine Parts · Slides 16–28

Design intent · projects and files · constrained sketches · standard part features

- Design begins with function: Translate load, motion, envelope, interface, safety and service needs into measurable acceptance criteria.
- The Inventor file ecosystem: IPT parts feed IAM assemblies; IPN presentations explain assembly; IDW/DWG drawings release manufacturing intent.
- Projects resolve references: An IPJ defines workspace, libraries, templates and Content Center locations so dependent files stay discoverable.
- Coordinate systems anchor intent: Origin planes, axes, points and user work features give sketches stable references that survive change.

- Sketch geometry is not design intent: Lines and arcs describe shape; geometric constraints describe behaviour; dimensions describe size and position.
- Fully constrained is predictable: Remove unintended degrees of freedom without over-constraining; test by dragging before leaving the sketch.
- Parameters make change deliberate: Meaningful names, equations and units turn dimensions into a controllable design system.
- Choose the right base feature: Extrude prismatic forms, revolve axisymmetric forms, sweep a profile along a path, and loft between sections.
- Placed features carry manufacturing logic: Hole, thread, fillet, chamfer, rib and shell encode operations and functional details.
- Feature order affects resilience: Build stable primary form first, functional cuts next, then finishing features such as fillets and chamfers.
- Standards are a design input: Select the drawing standard, units, projection convention, threads and tolerances before detailing.
- Validate before reuse: Check units, materials, sketches, feature warnings, mass properties and naming before a part enters an assembly.

Lab 1 — Design Brief to Parametric Base Part · Slide 27

Scenario: Create a controllable base component for a compact universal-joint guard and prove that the sketch captures the design brief.

Acceptance evidence: Sketch reports Fully Constrained and is centred on stable origin geometry.; Named parameters control the main dimensions without failed features.; The IPT opens from the lab project with no missing reference or update warning.; Evidence shows the requirement-to-feature rationale and a successful dimension change.

Topic 2 — Generate CAD Models · Slides 29–41

Advanced features · work geometry · freeform and sheet metal · model interoperability

- Analyse form before modelling: Classify the part as prismatic, rotational, path-driven, section-driven, freeform or sheet metal before choosing tools.
- Work geometry controls difficult features: Offset planes, tangent planes, axes and points create references where a natural face does not exist.
- Projected geometry links context: Project only the edges needed; excessive external dependencies make the feature tree fragile.
- Multi-body supports top-down design: Develop interacting solids in one context, then derive or make components while retaining intent.
- Patterns encode repetition: Rectangular, circular and path patterns reduce duplicated work and keep repeated geometry consistent.
- Mirrors encode symmetry: Mirror features or bodies across a stable plane instead of rebuilding symmetric geometry.
- Freeform solves organic transitions: T-spline forms suit ergonomic and aesthetic surfaces; validate curvature and convert deliberately.
- Sheet metal is rule-driven: Thickness, bend radius, relief and K-factor govern both folded geometry and manufacturable flat patterns.
- Orthographic views are a test: Front, top, side and isometric views expose missing geometry, asymmetry and unclear feature relationships.
- Imported CAD needs inspection: Confirm units, body type, geometry health and associativity before using STEP, IGES, SAT or Parasolid data.

- Direct edits are tactical: Move face, delete face and replace face solve local changes; parametric features remain preferable for design intent.
- Model health is evidence: A clean browser, resolved references, named features and stable update behaviour are part of the deliverable.

Lab 2 — Advanced Part, Sheet Metal and Orthographic Review • Slide 40

Scenario: Develop two alternative CAD models for a protective cover: a lofted solid concept and a manufacturable folded-sheet option.

Acceptance evidence: Loft sections are fully constrained and the resulting body has no visible discontinuity or failed feature.; Sheet-metal thickness and bend rule are consistent and the flat pattern is valid.; Orthographic views agree with the 3D models and show the intended geometry clearly.; The design comparison explains why one modelling method is preferred for production.

Topic 3 — Assembly Modeling • Slides 42–55

Top-down and bottom-up structures • degrees of freedom • joints and constraints

- Assembly structure communicates product logic: Parts, subassemblies and purchased items should mirror manufacturing, service and BOM expectations.
- Bottom-up reuses proven parts: Place independently designed components, then define their relationships and allowable motion.
- Top-down controls interfaces: Start from a layout or master geometry so interacting parts derive from shared functional references.
- Every component starts with six DOF: Three translations and three rotations remain until grounding, constraints or joints remove them.
- Constraints position geometry: Mate, flush, angle and insert relationships remove specific degrees of freedom between selections.
- Joints describe motion: Rigid, rotational, slider, cylindrical, planar and ball joints capture intended movement directly.
- Ground one stable reference: Ground the base component or use origin relationships so the mechanism has a predictable frame.
- Constraint order matters: Constrain primary locating surfaces first, secondary alignment next, then motion limits and detail relationships.
- Interference is a release blocker: Use collision and interference analysis to distinguish acceptable contact from overlapping solids.
- Representations reduce complexity: Design views, positional representations and model states support review, drawings and performance.
- Imported components need ownership rules: Decide whether external geometry remains associative, becomes a reference or is converted into a controlled part.
- Assembly drawings tell the build story: Views, parts lists, balloons and notes communicate order, identity and critical relationships.

Lab 3 — Bottom-Up Universal Joint Assembly • Slide 53

Scenario: Assemble the supplied universal-joint components from proven part files and retain the mechanism's intended rotation.

Acceptance evidence: All references resolve locally and the assembly opens without warnings.; The base is grounded, required components are constrained, and intended rotation remains possible.; Interference findings are reviewed and any unacceptable overlap is corrected.; The assembly drawing includes views, parts list and balloons that agree with the IAM BOM.

Lab 4 — Top-Down Layout and Assembly Relationships · Slide 54

Scenario: Control a universal-shaft variant from a master layout and compare that interface-driven approach with the bottom-up assembly.

Acceptance evidence: Named layout parameters control the key interfaces and update the assembly predictably.; No redundant or conflicting relationship remains in the browser.; Nominal and service positional representations are saved and demonstrably different.; The comparison correctly distinguishes top-down interface control from bottom-up reuse.

Topic 4 — Weldments · Slides 56–68

Preparation · weld features · post-weld machining · material and inspection decisions

- A weldment extends an assembly: It adds ordered preparation, weld and machining groups while retaining assembly relationships.
- Conversion is deliberate: A regular assembly can become a weldment, but the conversion is one-way and should follow a saved baseline.
- Preparation precedes joining: Chamfers, gaps and reliefs belong in the preparation group because they exist before the weld is applied.
- Weld features carry intent: Fillet, groove and cosmetic welds communicate joint type, extent and manufacturing expectation.
- Machining happens after welding: Post-weld holes, facing and cuts belong after the weld group when manufacturing sequence matters.
- Material choice affects the joint: Strength, weldability, corrosion, thermal response, availability and cost must be considered together.
- Standards remove ambiguity: Weld symbols, sizes, lengths, pitch, finish and process notes must follow the selected standard.
- BOM structure reflects fabrication: Decide whether the weldment is treated as one fabricated item or reported through its components.
- Distortion is a design risk: Heat input, restraint, sequence and access can change final geometry even when the CAD nominal is perfect.
- Inspection needs access: Design for visual inspection, gauges or NDT access where the risk and specification require it.
- Exploded views still need judgement: A weldment may require pre-weld and post-weld communication rather than a generic exploded assembly.
- Review manufacturability early: Check joint access, fit-up, tolerance stack, fixturing and post-process operations before release.

Lab 5 — Convert an Assembly to a Weldment · Slide 67

Scenario: Prepare a small fabricated bracket assembly for welding, inspection and post-weld machining using supplied assembly components.

Acceptance evidence: The working IAM is a weldment with correctly ordered preparation, weld and machining features.; Material choice is justified using at least four relevant criteria.; The drawing communicates weld type, extent and inspection intent under the selected standard.; Fabrication access and post-weld operations have been reviewed and recorded.

Topic 5 — Deliverables · Slides 69–81

Drawing views · dimensions and GD&T · parts lists · release-ready communication

- A drawing is a controlled communication: It translates design intent into a release package that manufacturing and inspection can interpret consistently.
- Templates set the contract: Sheet size, border, title block, units, projection, styles and revision fields should be correct before views are placed.
- Base view establishes orientation: Choose the most descriptive orientation, scale, hidden-line treatment and design representation.
- Projected views preserve alignment: Create orthographic and isometric views from the base so projection relationships stay associative.
- Sections explain internal form: Use cutting planes only where they reveal otherwise hidden geometry; hatch and labels must remain clear.
- Details enlarge critical areas: A detail view focuses tolerance, fit or feature complexity without crowding the parent view.
- Dimension for manufacture: Select functional datums and avoid duplicated, chained or ambiguous dimensions that hide tolerance accumulation.
- GD&T controls variation: Feature control frames define characteristic, tolerance zone, material condition and datum references.
- Datum schemes simulate function: Primary, secondary and tertiary datums constrain the part in the sequence used for assembly or inspection.
- Parts lists and balloons link identity: Item numbers, quantities, part numbers and descriptions must match the assembly BOM.
- Associativity needs checking: A green model update is not proof the drawing still communicates correctly; inspect every affected view and note.
- Release uses a checklist: Verify views, scales, units, tolerances, materials, notes, revision, approvals and exported deliverables.

Lab 6 — Production Drawing and GD&T Package · Slide 80

Scenario: Create a release-ready drawing for an engine-block component using views, dimensions, tolerances and a functional datum reference frame.

Acceptance evidence: Base, projected, section, detail and isometric views are correctly aligned and readable.; Dimensions are non-duplicated, trace to functional datums and avoid uncontrolled tolerance chains.; GD&T is syntactically complete and the datum scheme reflects assembly or inspection function.; The drawing updates associatively and passes the release checklist.

Topic 6 — Review Final Designs · Slides 82–95

Presentation and animation · material evidence · automation · design-for-manufacture review

- Review against requirements: A design is ready only when function, interfaces, performance, manufacture, safety and evidence meet the brief.
- Scenes preserve review viewpoints: Presentation scenes can show different representations, camera positions and communication purposes.
- Tweaks explain assembly: Explode direction, distance, order and trail visibility should reveal relationships without implying impossible motion.
- Storyboards organise time: A storyboard sequences component movement, camera actions and snapshots into a repeatable explanation.
- Animation is evidence, not decoration: Use movement to prove order, clearance or function; the static drawing must still carry complete meaning.
- Materials connect geometry to performance: Assign physical materials and verify density, mass, centre of gravity, appearance and manufacturing implications.

- Parameters expose design controls: Named user parameters make key dimensions visible to review and enable repeatable variants.
- iLogic captures rules: Rules can validate ranges, switch features, update properties and automate drawings when the intent is explicit.
- The API extends automation: Inventor's object model supports documents, components, features, properties and batch workflows; govern scripts as production tools.
- DFMA reduces lifecycle cost: Simplify part count, handling, fastening, tool access and process steps without compromising function.
- Change review is traceable: Record the issue, decision, affected files, evidence, owner and approval before releasing a revision.
- Productivity is system design: Templates, libraries, naming, model states, automation and checklists should reduce variation across the whole team.

Lab 7 — Exploded View, Storyboard and Animation • Slide 93

Scenario: Explain the assembly and service order of a small engine using an exploded presentation and a concise animation.

Acceptance evidence: All presentation references resolve and the source IAM remains unchanged.; Exploded paths are readable, non-colliding and consistent with a plausible assembly sequence.; Storyboard timing and snapshots communicate the same ordered narrative.; The exported output and drawing view are legible and suitable for stakeholder review.

Lab 8 — Material, Automation and Final Design Review • Slide 94

Scenario: Review an engine-block part against requirements, compare material options and encode a controlled design variant using named parameters and a simple rule.

Acceptance evidence: Review evidence addresses all stated requirements and clearly separates fact, assumption and recommendation.; Material decision uses multiple performance and manufacturing criteria with traceable mass-property evidence.; Named parameters and the rule control a valid range without failing the model.; The approved change, affected drawing and DFMA implications are recorded before release.

Resources Required

- Windows computer with Autodesk Inventor 2024 or later.
- Current slide deck, Learner Guide and complete labs folder.
- Projector and demonstration workstation.
- Access to the LMS for courseware, attendance and assessment submission.
- Assessment question papers and trainer-only answer keys held under the approved access controls.

Assessment

- Written Assessment (WA-SAQ): 8 open-ended questions, K1–K8, 60 minutes, open book.
- Practical Performance (PP): 6 tasks, A1–A11, 90 minutes, open book.
- Assessment block: Day 2, 3:30 PM–6:00 PM.
- Learners submit completed question papers on the LMS; answer keys remain trainer-only.